Beyond Static Reach: Learning to Wield a Flexible Tool from Sparse Visual Feedback

Giap Nguyen Trong[1], Son Dinh Bao[1], Long[2], Hai-Nguyen Nguyen[2,†], Chien Truong-Quoc[1,†]

***Abstract*—A flexible tool offers capabilities that its settled shape does not describe: transient motion can carry its tip beyond a measured quasi-static reach reference or bring it to a holdable target sooner. Exploiting this motion requires predicting the response to robot commands despite unobserved deformation and velocity. We present a learning-and-planning framework for transient endpoint control from sparse visual feedback. Tool-dependent excitation supplies data for a history-conditioned predictor of future observations under candidate commands. Receding-horizon planning then optimizes target entry and terminal behavior while leaving the intervening path free. The formulation accommodates different command interfaces and predictor architectures without requiring explicit tool-shape or material-parameter estimation. We evaluate one instantiation on a simulated single-joint arm carrying a 0.30 m flexible rod, observed through three visual landmarks, together with a separate physical-video prediction study using the same observation representation. On eight beyond-static targets located 30–118 mm from the nearest measured quasi-static position, the learned closed-loop controller enters all 40 evaluated episodes, whereas quasi-static control enters none. Prediction remains effective under real visual sensing, but reliable low-speed capture is not achieved consistently. These results establish controlled beyond-static entry in the evaluated setting while identifying terminal capture as the remaining limitation.**

***Index Terms*—Flexible-tool manipulation, dynamic manipulation, learned dynamics, model predictive control, sparse visual feedback, deformable-object manipulation.**

I. Introduction

A robot holding a flexible tool gains more than a geometric extension of its hand. During motion, the tool endpoint can pass through task-space positions that its settled shape does not describe. Such positions need not be holdable to be useful: an endpoint task may only require the tool to reach them transiently. Work on flexible-tool manipulation instead tends to treat deformation as something to be modeled and compensated [1], so useful reach is determined by positions the tool can hold after settling.

Using transient motion deliberately is a problem of partial observability. Hidden deformation, velocity, and flexible-mode phase affect where the endpoint moves next, so similar current observations can admit different futures. Rather than reconstructing the full flexible state, we ask whether a short observation history contains enough dynamic context to predict the endpoint response to a candidate robot command over a control horizon. A swept region alone is not a control capability; throughout the paper we therefore distinguish transient visitation from controlled beyond-static entry.

Related work. Dynamic nonprehensile manipulation exploits inertia and other transient effects to generate motion unavailable to quasi-static control [2], and related ideas have been applied to deformable-object planning [3], to goal-conditioned dynamic manipulation of deformable objects through iteratively refined actions [4], and to learned dynamic manipulation of flexible objects [5]; broader challenges are reviewed in [1]. Flexible-object dynamics can instead be identified from input–output measurements [6], and recent work uses brief exploratory motion to identify rope-specific physical parameters and then manipulates an unfamiliar rope dynamically with them [7]. Learned dynamics can support short-horizon sampling-based control [8], and information-theoretic MPC provides the sampling framework used here [9]. Our question is complementary: rather than identifying an unfamiliar tool or transferring across tool instances, we learn a direct finite-horizon response model for one robot–tool system with one actuated joint, define success at its observed tip, and ask what transient capability that makes deliberately available beyond a measured quasi-static reference.

Our approach learns that response from observation and command histories produced by tool-dependent excitation, which exposes both the driven and the free motion of the tool. The learned finite-horizon model is then rolled out inside receding-horizon planning to score candidate robot commands, so the tool's intermediate motion emerges from its own dynamics rather than being prescribed as a reference trajectory. Fitting is offline; online, new observations refresh both the history and the plan.

We study this question using a sparse visual observation representation shared by simulation and a separate physical-video study. Three tracked landmarks (elbow, hand, tip) are reduced to relative tip position and observed forearm angle; the learned model receives these derived quantities rather than camera images, joint state, curvature, strain, or simulator state. In simulation, the command interface is a joint-space trajectory instantiated as a commanded forearm angle tracked by a low-level servo. The physical study shares the observation representation but records no commanded angle and runs

*This research is funded by Hanoi University of Science and Technology (HUST) under project number T2025-PC-024.

[1]Giap Nguyen Trong, Son Dinh Bao and Chien Truong-Quoc are with the School of Mechanical Engineering, Hanoi University of Science and Technology, Hanoi, Vietnam chien.truongquoc@hust.edu.vn

[2]Long and Hai-Nguyen Nguyen are with the School of Science Engineering and Technology, RMIT University, Ho Chi Minh City, Vietnam hainguyen.nguyen@rmit.edu.vn

†Corresponding authors.

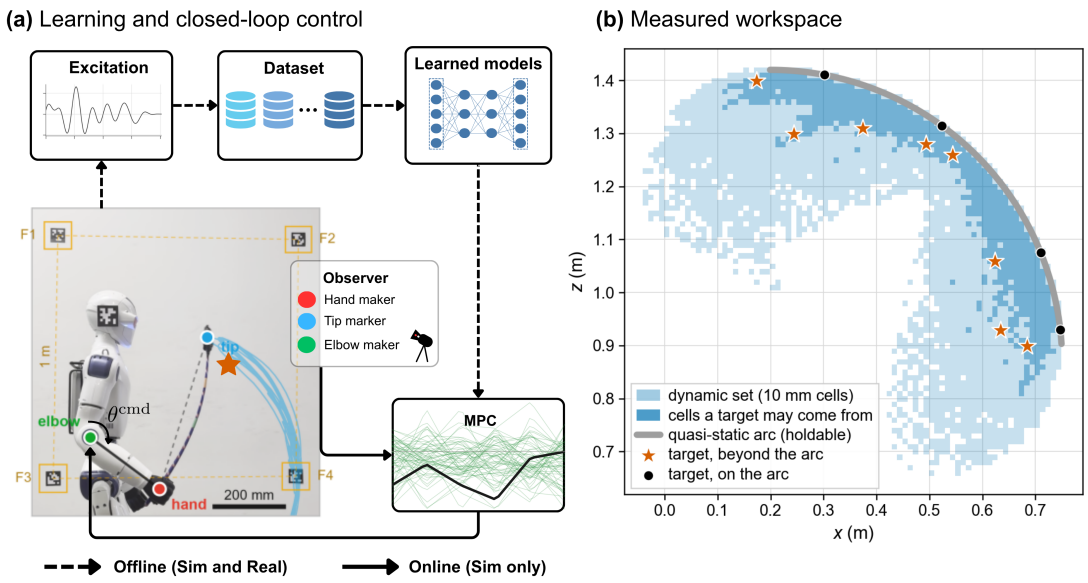


Fig. 1. System overview. **(a)** Offline excitation generates sparse visual observations for learning a finite-horizon endpoint response model. Online, the learned model is rolled out within sampling-based MPC to evaluate candidate commands, which are tracked by the low-level position controller. The simulation and physical-video studies share the same observation representation, while the command interface and closed-loop control are evaluated only in simulation. **(b)** Measured quasi-static reach reference and empirical transient occupancy in the sagittal x–z plane. Darker cells indicate well-sampled regions used for target selection. Occupancy denotes visitation, not holdability or controllability.

no controller. We ask three linked questions: whether short visual histories predict large transient motion, whether those predictions support repeatable control beyond the measured quasi-static reach reference, and whether successful entry also yields low-speed capture.

Contributions.

1) **A sparse visual response formulation.** We predict short-horizon endpoint response from sparse visual history and candidate robot commands without explicitly reconstructing tool deformation or material state.
2) **Finite-horizon prediction for online planning.** Observation-space models predict 200 ms of endpoint motion with millimetre accuracy at rollout rates compatible with receding-horizon control. The results also show that prediction depends strongly on whether the future input matches the command representation available to the planner.
3) **Controlled beyond-static entry and its boundary.** We separate transient visitation, deliberate entry, and low-speed capture, and show that learned-model predictive control can repeatedly access evaluated targets beyond the measured quasi-static reference while reliable terminal capture remains unresolved.

The rest of the paper follows that order. §II formalises the partially observed robot–tool setting, the sensing and command interfaces, and the two endpoint tasks. §III describes the elbow servo, the two corpora, the observation-space predictor, and the sampling-based planner. §IV reports the simulation and physical-video results, and §V–§VI state what they establish and what they leave open.

II. Problem Formulation

System. A simulated humanoid right arm holds a 0.30 m flexible rod modeled as a five-segment chain of torsional-spring hinges, a pseudo-rigid-body (PRB) approximation in the sense of [10], with a 0.1 kg tip mass in MuJoCo [11]. The shoulder is fixed, and the elbow is the only actuated robot joint. The physics step is 1 ms; sensing and control run at 120 Hz (frames of 8, 8, 9 ms).

Partially observed robot–tool dynamics. Let s_t denote the unobserved robot–tool state, including rod deformation and its rate, and let b specify the tool, grasp, boundary condition, and command interface. Then

$$s_{t+1} = F_b(s_t, u_{t+1}), \qquad o_t = G_b(s_t) + \eta_t, \tag{1}$$

where $u_{t+1} \in \mathcal{U}$ is the command applied between observations and o_t is the noisy measurement. Neither s_t nor F_b and G_b reaches the learned model or the controller.

Sensing. Three markers, at the elbow, the hand, and the rod tip, are projected through a virtual pinhole camera 1.55 m from the motion plane (0.939 mm/px on that plane, 1080×1920 px portrait frame) and perturbed by independent Gaussian noise of 1 px per marker per frame before metric conversion. The observation o_t of (1) is realised as the tip position relative to the hand in the plane, $x_t = p_t^{\mathrm{tip}} - p_t^{\mathrm{hand}} \in \mathbb{R}^2$, and the observed forearm angle $\theta_t = \operatorname{atan2}(p_t^{\mathrm{hand}} - p_t^{\mathrm{elbow}})$. The elbow marker sits 30 mm nearer the camera than the hand and tip markers, so θ_t carries the small perspective bias a real rig has. Neither the learned model nor the controller receives joint state, rod curvature, strain, PRB state, or other simulator state. Noise-free simulator quantities are used only for evaluation.

Command. The control input is a forearm-angle trajectory θ^{cmd}. A proportional–derivative elbow servo at the physics rate turns it into torque; the resulting torque is logged and never planned. The planner reconstructs an absolute tip for its cost from the observed elbow, the forearm length, and the *planned* angle, $\hat{p}^{\mathrm{tip}} = p^{\mathrm{elbow}} + L\,[\cos\theta^{\mathrm{cmd}}, \sin\theta^{\mathrm{cmd}}] + \hat{x}$, so no future measured hand or tip enters any rollout.

Static and dynamic workspaces. We define the quasi-static workspace as

$$\mathcal{W}_{\mathrm{qs}} = \left\{ p^{\mathrm{tip}}_{\infty}(\theta) : \theta \in \Theta \right\},$$

the set of settled tip positions under slow elbow motion over the declared joint domain Θ. It is an empirical reference measured over that domain, not a certificate that no equilibrium exists outside it. Dynamic reach depends additionally on the initial state and the available time. Writing P_b for the map extracting the tip position from a state, the finite-time dynamic reach is

$$\mathcal{W}_{\mathrm{dyn}}(s_0, T; b, \mathcal{U}, \Theta) = \bigcup_{\boldsymbol{\theta}^{\mathrm{cmd}}_{0:T} \in \mathcal{U}} \left\{ P_b(s_t) : t \in [0, T] \right\},$$

the task-space projection of the states reachable from s_0 under admissible command trajectories within the horizon. The experiments measure the empirical occupancy $\widehat{\mathcal{W}}_{\mathrm{dyn}}$ of the excitation corpus, a finite sampled subset of that set: membership denotes observed visitation and is neither exhaustive nor evidence of holdability or controllability.

Task 1 — beyond-static target entry. Given a target c outside $\mathcal{W}_{\mathrm{qs}}$, the objective is to minimise the closest approach $\min_{t \le T} \| p^{\mathrm{tip}}_t - c \|$ of the true tip within a window T of 2.0 s. That closest approach is the headline metric, compared pairwise between policies on identical (target, seed) episodes; entry, the same quantity falling within the 20 mm hit radius, is reported as a secondary count. The tip need not remain at the target.

Task 2 — timed transfer and capture. The robot moves from a settled point $A \in \mathcal{W}_{\mathrm{qs}}$ to another point $B \in \mathcal{W}_{\mathrm{qs}}$ within a time budget T. The path is unconstrained. We report two metrics:

- **hit:** $\min_{t \le T} \| p^{\mathrm{tip}}_t - B \| \le 20$ mm;
- **capture:** over the final 100 ms dwell window $\mathcal{D}$, the mean tip-to-target distance is at most 25 mm and the mean tip speed is at most 0.15 m/s.

These are mean dwell criteria, not pointwise bounds. Entry is transient visitation of the target region and capture is low-speed terminal behaviour: distinct capabilities, separated so that a settling arrival is not scored by a transient pass. Neither task constrains the intervening path.

III. Method

Fig. 1(a) summarizes the pipeline. Offline, excitation produces the visual observation sequences that train the finite-horizon response models; in simulation it is applied as forearm-angle commands through the elbow servo at 120 Hz. Online, the vision-derived relative tip and forearm angle reach the sampling-based planner, which proposes a forearm-angle trajectory that the servo tracks.

A. Elbow Servo

The elbow servo is fixed before model training and is not retuned: $k_p = 400$ N m/rad and $k_d = 10.65$ N m s/rad. Using the measured elbow inertia $I = 0.0709$ kg m^2, we set $k_d = 2\zeta\sqrt{k_p I}$ with $\zeta = 1$, corresponding to nominal critical damping under a local single-DOF approximation; this does not characterize the damping of the full arm–rod system. The closed loop reaches 90% of a 0.30 rad step in 58 ms, within one quarter of the measured first-mode period (400 ms), and has an amplitude ratio of 0.95 at f_1. On recorded forearm-angle trajectories, it achieves 0.066 rad tracking RMSE with 3.9% torque saturation. The gains were fixed before evaluating any task-level metric.

B. Data and Excitation

Two corpora share the same observation representation. Only the simulation corpus is commanded and used for control.

Simulation corpus. Excitation spans 1.25–4.00 Hz, covering the measured first mode at $f_1 = 2.500$ Hz with damping ratio $\zeta = 0.0829$ ($R^2 = 0.836$ over 10 peaks). Data are generated only when f_1 lies within this band. All 528 episodes are retained; none reaches a joint stop under the bounded angle commands. Each episode lasts 3.0 s (360 frames). Splitting is performed by episode to prevent leakage between overlapping windows. The train/validation/test split is 370/79/79 episodes, corresponding to 38,110/8,137/8,137 windows at stride three.

Physical-video corpus. A separate recording tests whether the same vision-derived endpoint dynamics remain learnable under real sensing. A humanoid arm swings an attached flexible rod in a plane approximately parallel to the image plane, recorded at ≈120 fps over 46 trials (Fig. 1(a)). Elbow, hand, and tip markers are tracked relative to four static fiducials spanning a known 1 m square. Those fiducials sit on the background wall, 2.35 m from the camera, while the arm swings 1.55 m from it, so their 1.423 mm/px scale overestimates the motion plane's by 1.52×; the metric scale is therefore taken as 0.939 mm/px by rescaling on the depth ratio. It is cross-checked against known lengths, which agree to +1.8% on a 60 mm tag side and +3.3% on a 230 mm rigid link, with 1.21 mm frame-to-frame tip variability (Fig. 2(d)); these check consistency and are not an independent calibration. The simulation's virtual camera is declared at that recovered scale and depth (§II), so the two corpora share the observation representation in metric terms and not only in form.

The observation and the prediction window are those of §II and §III-C: 250 ms of history predicting 200 ms ahead, matching the simulation's window in physical time. **The command side is not shared.** No commanded angle is recorded on this rig, so models train on the *measured* future angle, and no controller is run on it. A clip-level split of 31/7/8 trials yields 10,650/2,559/2,804 windows.

C. Observation-Space Dynamics Model

All models share the finite-horizon mapping

$$f_\phi : \left(x_{t-k+1:t}, \theta_{t-k+2:t}, \theta^{\mathrm{cmd}}_{t+1:t+H}\right) \rightarrow \hat{x}_{t+1:t+H}, \quad (2)$$

with $k = 30$ frames (250 ms) of relative-tip and measured-angle history and $H = 24$ steps (200 ms) of commanded future angle, matching the planning horizon. The history stands in for the hidden state of (1): recent relative-tip positions carry the direction and flexible-mode phase that a single frame omits, and the measured past angle separates driven from free response. The history control is the *measured* angle and the future control the *commanded* angle, which is what a planner can supply; the measured future angle is kept beside it only as a diagnostic score of the same checkpoint. We compare GRU [12], MLP, Koopman [13], and LNN [14] predictors, together with persistence and constant-velocity references. All learned models use the same horizon curriculum and validation-based early stopping.

D. Model Selection

Accuracy and rollout latency select the model jointly: a model is control-usable only if a planner-sized batch of 256 rollouts over 24 steps fits within the 25 ms replanning interval (three frames). The GRU attains the lowest held-out error *in simulation, under the commanded future it is planned with*, and completes that batch in 8.3 ms, so it is used for control; the LNN fits the interval with less margin, and the MLP and Koopman models are faster and less accurate (Table I, Fig. 3(a)). That qualification is examined in §IV-B.

E. Sampling-Based Predictive Control

At each replan, the controller samples 256 forearm-angle trajectories as six-knot splines (knot spread 0.15 rad, slew limited to 8 rad/s within a measured command box), rolls them out with f_ϕ over the 24-step horizon, evaluates the task cost on the reconstructed tip, and applies the first three commands (25 ms) before replanning from new marker observations. The sampling update follows information-theoretic MPC [9] with a population-scaled softmin temperature and without the path-integral control-cost cross term; we refer to it as sampling-based MPC. The softmin temperature is calibrated by a rule fixed before the sweep was read, which takes the largest temperature at which the executed (softmin-averaged) plan's own predicted closest approach stays within 5 mm of the best sampled candidate's, so that the update commits to the swing the population found instead of averaging it away. It was applied to the Task-1 evaluation grid itself, the same twelve targets and five seeds reported in §IV-D, so with no separate calibration split the adopted value, $T = 0.2$, is not statistically independent of the Task-1 numbers reported there. The supplement gives the swept belief gap and effective sample size.

For Task 1, the cost minimizes the closest tip-to-target distance over the horizon, with additional approach-time, command-deviation, smoothness, and constraint penalties. Task 2 adds terminal position, terminal velocity, dwell, and deadline-progress terms. Neither task imposes a reference path.

Reference policies. Open-loop learned shooting uses the same GRU, task cost, 24-step horizon, six-knot spline representation and 256 candidates, but plans once and executes blind, chaining consecutive horizon-length blocks through its own predicted observations rather than through new marker data. It is not a strict feedback ablation: the MPC perturbs a warm-started nominal with Gaussian knot noise and applies a softmin-weighted average of the population, whereas the shooter samples the command box uniformly and applies its single best candidate. The comparison is therefore between closed-loop replanning and open-loop learned shooting, not an isolation of feedback. The quasi-static policy slews the commanded angle along the measured static map without exciting the rod, and the hold policy keeps the current angle. For Task 2, the fast-kinematic policy is handed the target's measured static forearm angle and drives to it on a deadline-scaled minimum-jerk trajectory; it is a feasibility reference for low-speed arrival, not an equal-information competitor to the learned planner, which receives only the target position. All references are tracked by the same servo.

IV. Experiments and Results

Figs. 1(a,b)–5 evaluate the framework in simulation, using five seeds per control condition unless stated otherwise. A separate physical-video dataset evaluates the same observation representation under real sensing. Its acquisition and processing are described in §III-B, and its held-out prediction results are reported alongside the simulation results in §IV-B. No closed-loop hardware control is evaluated.

A. Transient Motion Beyond the Quasi-Static Reference

Both measurements use the same commanded elbow range, $[-0.72, +0.72]$ rad. A slow sweep yields an 832 mm quasi-static arc from 241 settled tip positions. Under the same boundary condition, dynamic excitation produces empirical tip occupancy extending up to 411 mm beyond this reference (95th percentile: 313 mm), with an occupied area 5.2 times that of a 30 mm band around the arc. These values describe positions visited under the tested excitation family; they neither exhaust the finite-time dynamic workspace nor rule out other settled poses.

Targets are drawn only from well-sampled occupancy. A beyond-static target must lie in a 10 mm cell visited at least 20 times with all eight neighbouring cells also visited; on-arc controls are selected from settled nodes crossed by the dynamic trajectories. Of 2,597 occupied cells, 615 satisfy this screen. Fig. 1(b) shows eight beyond-static targets 30–118 mm from the nearest settled tip position and at least 50 mm apart, together with four on-arc targets spaced by at least 150 mm.

B. Prediction of Large Transients

Fig. 2 evaluates prediction across the large endpoint excursions relevant to beyond-static reaching. In simulation, the GRU achieves 2.55 mm full-horizon RMSE over 200 ms,

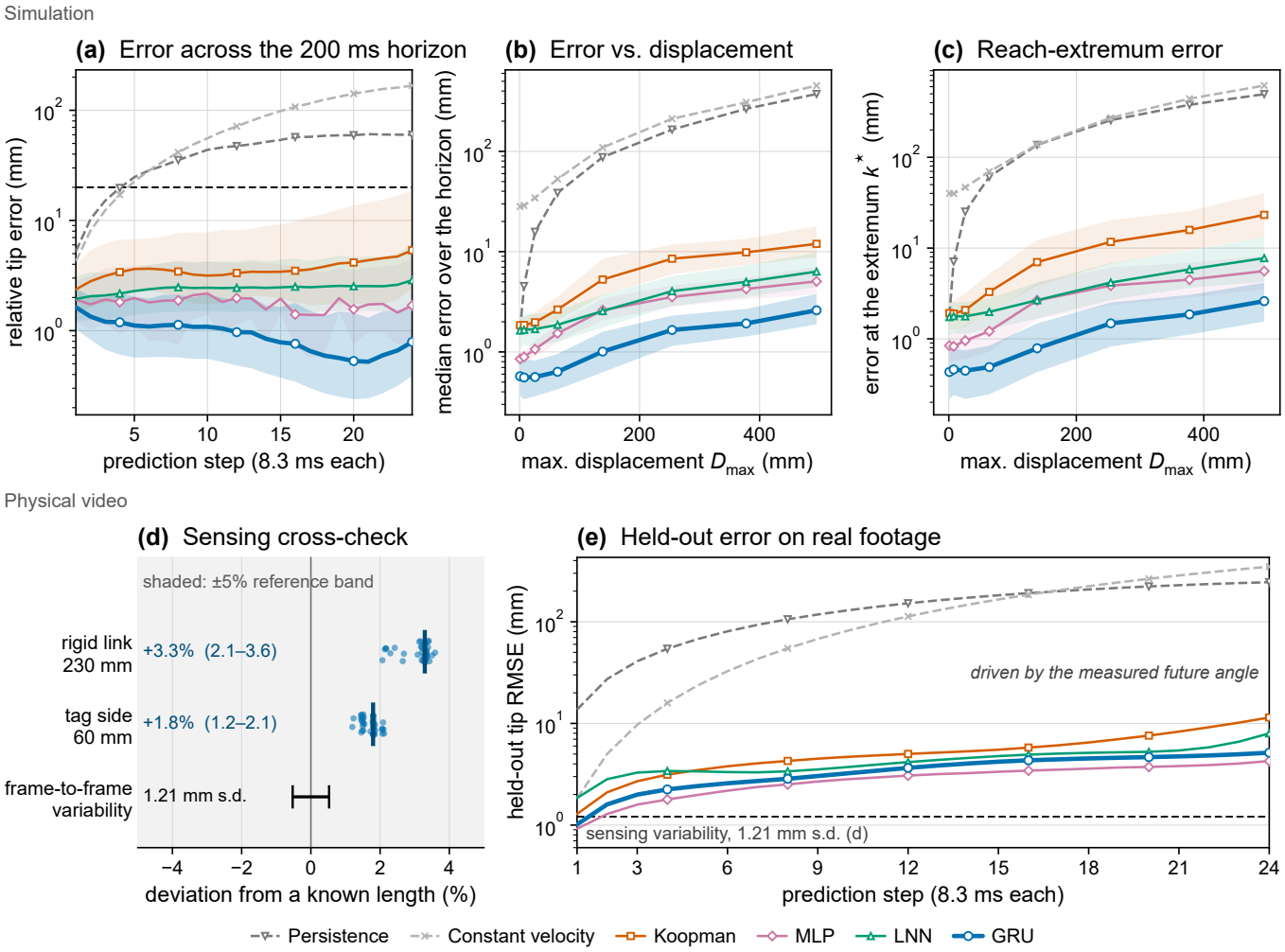


Fig. 2. Held-out prediction in simulation and on physical video over the same 200 ms horizon using the same six predictor families. The two evaluations use separate corpora and independently trained models; each recovers its own metric scale, and the simulation camera is declared at the scale recovered from the physical rig. **(a)–(c) Simulation:** the MuJoCo flexible-rod plant, 8,137 windows from 79 held-out episodes; lines show medians, with 25–75% bands drawn for the learned models. $D_{\max}$ denotes the maximum true relative-tip displacement over the horizon, grouped into eight equal-count bins. **(a)** Error across the 24-step rollout; the dashed line marks the 20 mm Task 1 hit radius for scale. **(b)** Median horizon error versus $D_{\max}$. **(c)** Error at the true reach extremum $k^\star$ versus $D_{\max}$. **(d)–(e) Physical video:** 46 trials at ≈120 fps, 2,804 held-out windows from 8 unseen clips (§III-B; rig in Fig. 1(a)). **(d)** Length cross-checks against a 230 mm rigid link and a 60 mm AprilTag side over all 46 trials, together with frame-to-frame tip variability; these assess metric consistency but are not an independent calibration. **(e)** Held-out tip RMSE at each prediction step under the measured future angle; the dashed level shows the variability from (d) for scale. One training seed is used for the physical study, and no closed-loop hardware control is evaluated.

compared with 170.7 mm for persistence, with 2.34 mm error at the final step and 2.73 mm at the true reach extremum. Its error grows only 1.07× from the first to the last step. Removing the synthetic marker noise reduces RMSE from 2.55 to 1.55 mm.

Each model rolls out 200 ms of tip motion conditioned on the forearm-angle trajectory over that horizon, and two versions of that trajectory exist: the angle commanded to the servo, which a planner knows because it is choosing it, and the angle the arm actually reaches, which the servo tracks to a median 0.060 rad RMSE. Models are trained on the commanded angle. Scoring the same checkpoints on the measured angle instead, with no retraining, raises the GRU from 2.55 to 22.63 mm, a factor of 8.9, and leaves every learned model between 20.8 and 23.6 mm (Table I). The substitution costs more accuracy than any difference between the models, so the two angle signals are not interchangeable inputs. Primary errors are reported under the commanded angle because that is the one a planner supplies.

The physical-video study shows the same sensitivity with the roles reversed. That rig records no command, so its models are trained on the measured future angle and reach 3.05–6.01 mm full-horizon RMSE over 2,804 held-out windows, against 163.40 mm for persistence and 178.66 mm for constant velocity (Fig. 2(e)). Substituting a counterfactual angle reconstructed under a fixed-base assumption, the closest available stand-in for what a planner would supply on this rig, raises the learned models to 9.16–10.39 mm and reverses their order: the MLP falls from first to last and the LNN rises from third to first. Together the two studies say that a predictor must be

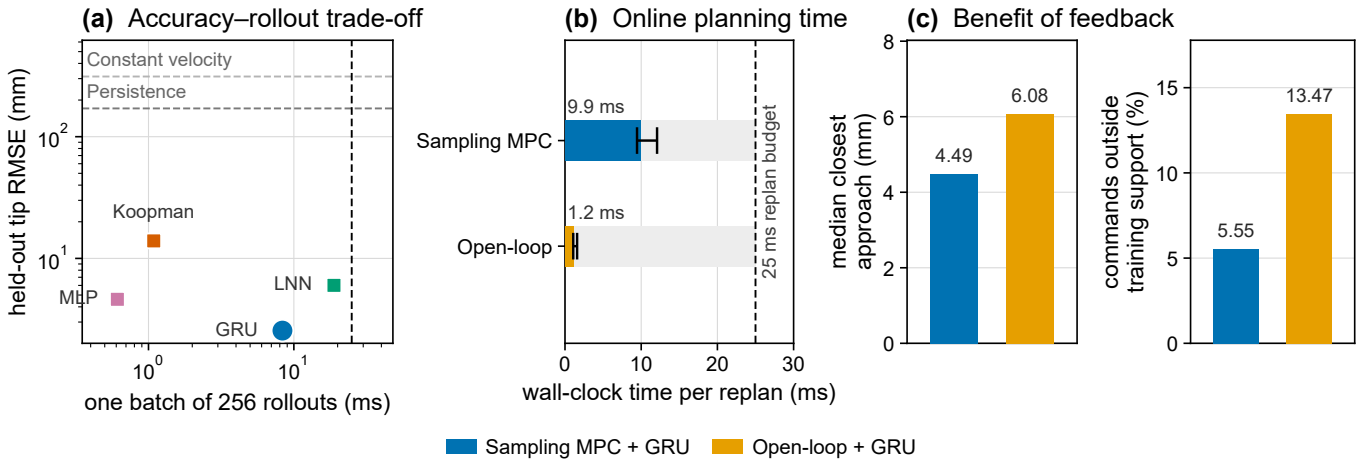


Fig. 3. Prediction accuracy, rollout latency, and closed-loop versus open-loop control. **(a)** Held-out tip RMSE versus latency for 256 rollouts; the dashed line marks the 25 ms replanning budget. **(b)** Replanning time over the 40 beyond-static Task 1 episodes; bars show medians and caps the observed range. **(c)** Closed-loop replanning and open-loop shooting compared by median closest approach and the fraction of commands outside the training range. The closed loop is closer on 72% of matched $(\text{target}, \text{seed})$ pairs.

TABLE I
SIMULATION PREDICTION ACCURACY AND ROLLOUT LATENCY

Model	Tip RMSE (mm)	Extremum (mm)	Measured future RMSE (mm)	Batch (ms)
GRU (selected)	**2.55**	**2.73**	22.63	8.34
MLP	4.61	5.24	23.59	0.61
LNN	6.00	6.88	22.35	18.87
Koopman	13.93	21.32	20.77	1.09
Persistence	170.72	245.15	170.72	—

Primary errors use the commanded future angle; "Measured future" reevaluates the same checkpoints with the measured future angle. Batch latency is for 256 rollouts over 24 steps on an idle CPU.

TABLE II
BEYOND-STATIC TARGET ENTRY (8 TARGETS × 5 SEEDS)

Policy	Hit @20 mm	Closest (mm)	Hit @10 mm
sampling-based MPC	**40/40**	**4.49**	**37/40**
open-loop shooting	38/40	6.08	28/40
quasi-static	0/40	71.4	0/40
hold	0/40	268.8	0/40

Closest approach is the per-episode median over the 40 beyond-static episodes. Paired on identical (target, seed) episodes, the MPC is the closer of the two in 29 of 40 pairs (72%), by 4.0 mm at the median.

trained and scored on the same future-input representation it will be given.

C. Prediction and Closed-Loop Control

The selected GRU is fast enough for online planning: a batch of 256 rollouts takes 8.34 ms, and the realised closed loop requires 9.96 ms per replan at the median and 12.08 ms at most, within the 25 ms budget (Fig. 3(a,b)).

Fig. 3(c) compares closed-loop replanning with open-loop learned shooting. Both achieve high beyond-static entry rates (40/40 and 38/40 at 20 mm), but the closed-loop strategy is more precise: its median closest approach is 4.49 mm versus 6.08 mm, and it is closer on 29 of 40 matched $(\text{target}, \text{seed})$ pairs (72%). It also executes fewer commands outside the training range (5.5% versus 13.5%). Thus, in this comparison, closed-loop planning primarily improves precision and keeps execution closer to the model's training regime rather than expanding the observed reach.

D. Controlled Beyond-Static Entry

Sampling-based MPC enters all 40 beyond-static episodes with a 4.49 mm median closest approach, whereas the quasi-static policy enters none and remains 71 mm away at the median (Fig. 4, Table II). Every seed reaches every evaluated target, including the one 118 mm from the nearest settled tip position. The same quasi-static policy reaches all 20 on-arc episodes, showing that it remains effective on the measured quasi-static reference.

A 400 ms recursive rollout reduces the median closest approach to 2.92 mm, but extends the 200 ms predictor beyond its trained horizon, where held-out error rises from 2.55 to 4.68 mm. We therefore retain the 200 ms horizon for the primary results; other planner ablations are reported in the supplement.

Each episode is checked after execution for joint stops, contacts, torque saturation, diverged plans, and lost markers. No flag is raised on any of the 40 MPC episodes, and commanded-angle tracking has a median RMSE of 0.073 rad. They are diagnostics, not guarantees.

E. Timed Transfer and Capture

Task 2 tests whether successful entry also yields low-speed capture at statically reachable targets. Across seven time budgets (0.25–2.0 s), the sampling-based MPC reaches 256/280 episodes but captures only 84/280 (Fig. 5). Open-loop shooting reaches 273/280 and captures none, while the fast-kinematic reference reaches all 280 and captures 185/280. Because this reference is given the target's measured static angle, it serves only as a feasibility reference for capture.

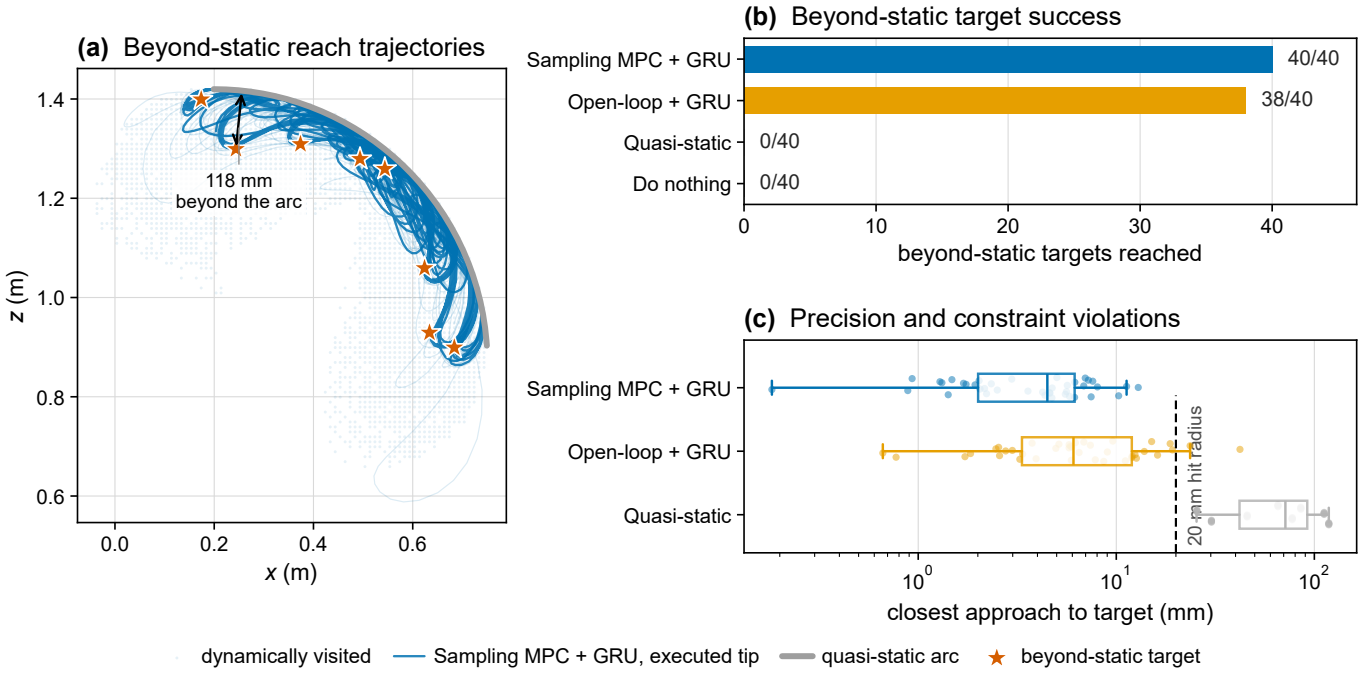


Fig. 4. Controlled beyond-static entry under learned-model planning. **(a)** Measured quasi-static arc, dynamically swept occupancy, eight beyond-static targets, and executed sampling-based MPC trajectories. The targets lie 30–118 mm from the nearest settled tip position; the arrow marks the most distant one. **(b)** Entries within the 20 mm hit radius over 8 targets × 5 seeds per policy. Sampling-based MPC enters all 40 episodes with a 4.49 mm median closest approach. **(c)** Closest approach for each episode. Constraint flags are diagnosed after execution rather than enforced during planning; no flag is observed in the 40 MPC episodes.

For the MPC, the median dwell distance is 12.1 mm, within the 25 mm capture bound, whereas the median dwell speed is 0.227 m/s, above the 0.15 m/s threshold. Thus, residual speed is the dominant failing criterion after entry. The 200 ms planning horizon is shorter than the longer task deadlines, and the capture rate remains 30–48% from $T = 0.6$ s onward; additional time therefore improves entry more reliably than settling.

V. Discussion

The measured transient occupancy should not be interpreted as controlled reach. Its 411 mm maximum excess records visitation under the tested excitation, whereas the controlled evaluation extends only to the furthest screened target, 118 mm from the nearest settled tip position. Because target selection requires prior occupancy support, this distance characterizes the evaluated target set rather than a discovered control boundary.

History conditioning addresses the partial observability of transient flexible-tool motion: similar endpoint observations can correspond to different hidden deformation and velocity, while a short history provides additional dynamic context without reconstructing the full flexible state. The prediction results show that this representation remains accurate over large excursions, but do not establish that the finite history resolves the underlying state. The physical-video study further supports vision-derived prediction under real sensing, but not controller transfer: the two studies share the observation representation, not the command interface, and all closed-loop results remain in simulation.

VI. Limitations

Simulation scope. Closed-loop evaluation is limited to one launch pose, one actuated joint, one flexible tool, one servo setting, one synthetic noise level, and one training seed. The pixel noise is not a measured tracking residual, and the virtual sensing pipeline does not model marker loss. Constraint violations are diagnosed only after execution. The softmin temperature is calibrated on the Task 1 evaluation episodes themselves, so those results are not independent of that setting. Reported planning times establish feasibility against the 25 ms budget but are hardware dependent.

Mechanism and generalization. The experiments do not isolate elasticity from inertial effects, and actuator bandwidth is not varied. The capture results identify residual tip speed as the dominant failing criterion but do not establish its physical cause. No architecture- or controller-level optimality is claimed.

Physical-video study. The physical study comprises one 46-trial recording session and one training seed. Models use the measured future angle because no commanded angle is recorded, and no closed-loop hardware control is evaluated. Metric scale is cross-checked rather than independently calibrated, and prediction error is measured against tracked visual landmarks rather than independent tip ground truth. The rigid-base future input is a counterfactual reevaluation of the recorded trajectories, not a recorded command.

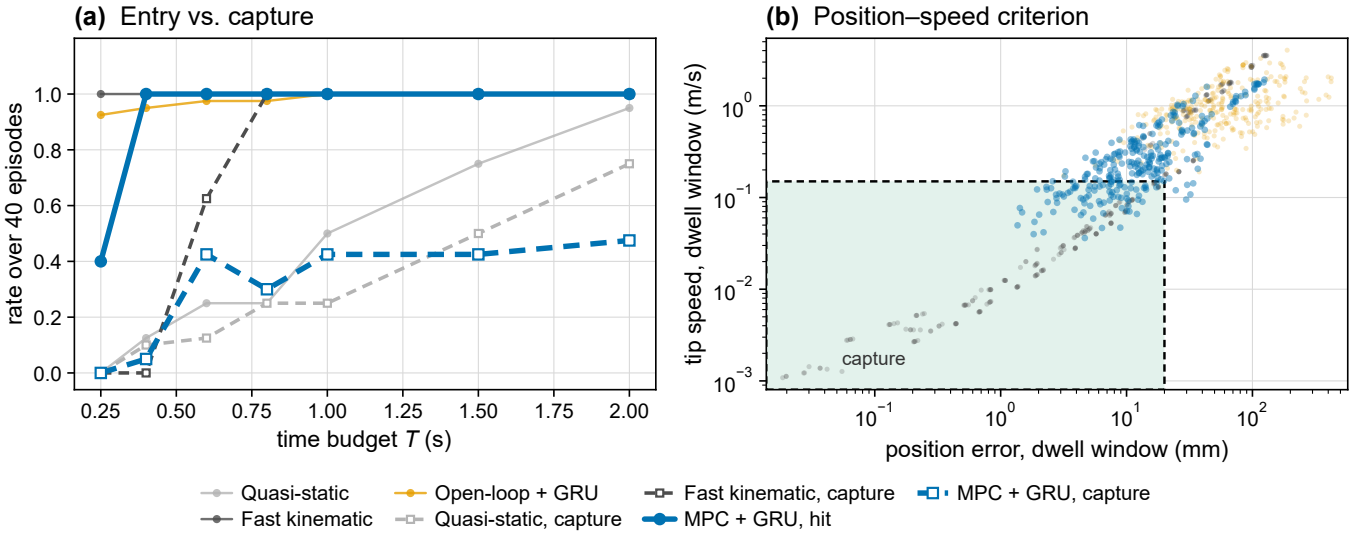


Fig. 5. Timed transfer and capture. **(a)** Task 2 hit and capture rates versus the time budget over 8 targets × 5 seeds; the servo-tracked fast-kinematic reference captures every episode from $T = 0.8$ s and 185 of 280 pooled, the sampling-based MPC 84 of 280, and open-loop shooting none. **(b)** Mean tip-to-target distance and mean tip speed over the final 100 ms dwell window; the shaded region satisfies both capture conditions, and the sampling MPC ends inside it in 84 of its 280 episodes.

VII. Conclusion

This work shows that transient flexible-tool motion can be treated as a controllable task-space resource rather than only as deformation to be suppressed. By learning short-horizon endpoint response from sparse visual history and candidate commands, the proposed framework enables deliberate access beyond a measured quasi-static reach without reconstructing the full flexible state or prescribing an intermediate path.

The results also distinguish transient visitation, controlled entry, and low-speed capture as separate capabilities. Prediction remains viable under real visual sensing, but closed-loop hardware control and reliable terminal capture remain open problems. Extending this framework to richer robot–tool systems therefore requires improving terminal regulation and robustness while preserving the benefits of transient motion.

References

[1] J. Zhu, A. Cherubini, C. Dune, D. Navarro-Alarcon, F. Alambeigi, D. Berenson, F. Ficuciello, K. Harada, J. Kober, X. Li, J. Pan, W. Yuan, and M. Gienger, "Challenges and outlook in robotic manipulation of deformable objects," *IEEE Robotics & Automation Magazine*, vol. 29, no. 3, pp. 67–77, Sep. 2022. [Online]. Available: http://dx.doi.org/10.1109/MRA.2022.3147415

[2] K. M. Lynch and M. T. Mason, "Dynamic nonprehensile manipulation: Controllability, planning, and experiments," *The International Journal of Robotics Research*, vol. 18, no. 1, pp. 64–92, Jan. 1999. [Online]. Available: http://dx.doi.org/10.1177/027836499901800105

[3] S. Zimmermann, R. Poranne, and S. Coros, "Dynamic manipulation of deformable objects with implicit integration," *IEEE Robotics and Automation Letters*, vol. 6, no. 2, pp. 4209–4216, Apr. 2021. [Online]. Available: http://dx.doi.org/10.1109/LRA.2021.3066969

[4] C. Chi, B. Burchfiel, E. Cousineau, S. Feng, and S. Song, "Iterative residual policy: For goal-conditioned dynamic manipulation of deformable objects," *The International Journal of Robotics Research*, vol. 43, no. 4, pp. 389–404, 2024. [Online]. Available: http://dx.doi.org/10.1177/02783649231201201

[5] K. Kawaharazuka, T. Ogawa, J. Tamura, and C. Nabeshima, "Dynamic manipulation of flexible objects with torque sequence using a deep neural network," in *2019 International Conference on Robotics and Automation (ICRA)*. IEEE, May 2019, pp. 2139–2145. [Online]. Available: http://dx.doi.org/10.1109/ICRA.2019.8793513

[6] M. Floren, S. Mamedov, J.-P. Noël, and J. Swevers, "Identification of deformable linear object dynamics from input-output measurements in 3d space," *IFAC-PapersOnLine*, vol. 58, no. 15, pp. 468–473, 2024. [Online]. Available: http://dx.doi.org/10.1016/j.ifacol.2024.08.573

[7] A. Jakobsson, A. Mahajan, K. Pullalarevu, K. Suresh, Y. Yao, Y. Mao, B. Duisterhof, S. N. Syed, and J. Ichnowski, "Wiggle and go! system identification for zero-shot dynamic rope manipulation," 2026, arXiv:2604.22102 [cs.RO]. [Online]. Available: https://arxiv.org/abs/2604.22102

[8] K. Chua, R. Calandra, R. McAllister, and S. Levine, "Deep reinforcement learning in a handful of trials using probabilistic dynamics models," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

[9] G. Williams, N. Wagener, B. Goldfain, P. Drews, J. M. Rehg, B. Boots, and E. A. Theodorou, "Information theoretic mpc for model-based reinforcement learning," in *2017 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, May 2017, pp. 1714–1721. [Online]. Available: http://dx.doi.org/10.1109/ICRA.2017.7989202

[10] L. L. Howell and A. Midha, "A method for the design of compliant mechanisms with small-length flexural pivots," *Journal of Mechanical Design*, vol. 116, no. 1, pp. 280–290, Mar. 1994. [Online]. Available: http://dx.doi.org/10.1115/1.2919359

[11] E. Todorov, T. Erez, and Y. Tassa, "Mujoco: A physics engine for model-based control," in *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, Oct. 2012, pp. 5026–5033. [Online]. Available: http://dx.doi.org/10.1109/IROS.2012.6386109

[12] K. Cho, B. van Merrienboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using rnn encoder–decoder for statistical machine translation," in *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 2014, pp. 1724–1734. [Online]. Available: http://dx.doi.org/10.3115/v1/D14-1179

[13] M. Korda and I. Mezić, "Linear predictors for nonlinear dynamical systems: Koopman operator meets model predictive control," *Automatica*, vol. 93, pp. 149–160, Jul. 2018. [Online]. Available: http://dx.doi.org/10.1016/j.automatica.2018.03.046

[14] M. Cranmer, S. Greydanus, S. Hoyer, P. Battaglia, D. Spergel, and S. Ho, "Lagrangian neural networks," 2020, arXiv:2003.04630 [cs.LG]; presented at the ICLR 2020 Deep Differential Equations Workshop.